

LEONARD THOMAS AARONS-DITSON

AI ENGINEER

Agentic Systems · Applied AI · Evaluation Infrastructure · Python/ML
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PROFESSIONAL SUMMARY

UCL-trained physicist and AI Engineer combining commercial data-science experience with several years of independent research and engineering. Builds AI agents, LLM evaluation infrastructure, structured-output pipelines and ML workflows in Python, with a record of deterministic validation, trace auditing and measurable results.

EDUCATION

University College London — Physics | PGCert Physics with Distinction (2021) · BSc Physics (2018)

Selected postgraduate results: **Advanced Quantum Theory 93.5%** · **Modelling Quantum Many-Body Systems 86%**. Postgraduate quantum-information research (2018–2021): computational modelling of spin-chain communication and open quantum systems in Python/Qiskit.

INDEPENDENT AI RESEARCH & ENGINEERING | 2023 – PRESENT

Coding-Agent Cross-Review — Multi-agent developer tooling (2026)

- Architected a dual-agent software-engineering workflow in which Claude Code and Codex CLI solve tasks in isolated Git worktrees and exchange schema-validated JSON reviews behind deterministic test gates (150 offline tests, ruff, strict mypy); in a live run the cross-review caught a Unicode lowercasing defect that four passing acceptance tests had missed.

Grounded Agent Planning Pipeline — Stateful agents, structured outputs, deterministic validation (2026)

- Engineered a stateful structured-output agent pipeline (Codex CLI JSON Schema, Pydantic validation, persistent run records with retries and idempotency controls) whose deterministic validators rejected unsupported claims in 151/251 initial plans across 259 runs on 257 real inputs, forcing evidence-grounded replanning.

Agentic Physics Bench — LLM evaluation, tool use, trace auditing (2026)

- Built a frozen, hash-verified LLM evaluation harness (162 scored episodes, deterministic grading, per-call trace controls) measuring when a model delegates to a numerical tool: optional tool use rose from 0/12 to 54/54 across versions with accuracy at ceiling, and a trace audit caught an undeclared second model in 31/48 development calls.

FleetCast — ML forecasting pipeline and feature engineering (2026)

- Developed a chronological demand-forecasting pipeline on public NYC taxi data (DuckDB, past-only feature engineering, leakage tests, Poisson gradient-boosted trees, Streamlit error-analysis dashboard), cutting holdout MAE 24% versus persistence across 13,440 predictions.

COMMERCIAL EXPERIENCE

Data Scientist | Sticky | London | Oct 2020 – Feb 2021

- Built Python statistical analyses and predictive ML models across customer, payment and product-placement data, contributing to a verified 27% increase in profitability.
- Built automated product-placement decision tools and investigated SaaS technical and operational failures, translating findings into recommendations for product and non-technical stakeholders.

Additional experience: NHS operational roles — East of England Ambulance Service (999 call handling, 2026) and Imperial College Healthcare (laboratory operations, 2023–2024) · Data Analyst, BPL Global (2017) · Project & Data Support, Canary Wharf Contractors (2016; Employee of the Month).

TECHNICAL STACK

AI / Agents: LLM APIs (Anthropic, OpenAI), agent workflows, multi-agent orchestration, tool use, structured outputs (Pydantic/JSON Schema), evaluation harnesses, trace auditing, Claude Code, Codex CLI, context engineering.

Engineering: Python, FastAPI, Pydantic, REST/JSON APIs, OAuth 2.0, Git/GitHub, Git worktrees, pytest, ruff/mypy, GitHub Actions/CI, SQL, PostgreSQL, SQLite, DuckDB, TypeScript/React, Streamlit.

ML / Data: NumPy, pandas, SciPy, scikit-learn, gradient-boosted trees, feature engineering, forecasting and time-series evaluation, Kalman filtering, linear programming, policy-gradient RL, gradient-based optimisation.

Scientific / HPC: C/C++, OpenCL, GPU/HPC computing, Jupyter, Qiskit, QuTiP, numerical modelling, simulation engineering, Monte Carlo (Brownian/GBM) simulation, Black–Scholes pricing.

Distinctions: Cumberland Lodge selection (1 of 15 UCL Physics students) · Competitive UCL rowing (Cambridge Head winners) · Organised a 72-hour rowathon raising £25,000 for Cancer Research UK.